

Preface

Test

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Last update: Test 2026 September 25

Chapter 1

Banach Algebras

§ 1.1. Basic Definitions and Results

Definition 1.1.1. An *algebra* over $\mathbb{F}$ is a vector space $\mathcal{A}$ over $\mathbb{F}$ that has a multiplication operation making $\mathcal{A}$ into a ring such that, for any $\alpha \in \mathbb{F}$ and $a, b \in \mathcal{A}$, we have $\alpha(ab) = (\alpha a)b = a(\alpha b)$. If $\mathcal{A}$ admits an identity element, we say $\mathcal{A}$ is *unital*.

Definition 1.1.2. A *normed algebra* is an algebra $\mathcal{A}$ over $\mathbb{F}$ which has a norm $\|\cdot\|$ making $\mathcal{A}$ into a normed linear space and also admitting the property that, for any $a, b \in \mathcal{A}$, we have $\|ab\| \leq \|a\| \|b\|$. If $\mathcal{A}$ is a Banach space, then $\mathcal{A}$ is called a *Banach algebra*. If $\mathcal{A}$ has an identity e , then it is assumed that $\|e\| = 1$.

Proposition 1.1.1. If $\mathcal{A}$ is an algebra without identity, define algebraic operations on $\mathcal{A} \times \mathbb{F}$ by:

- (i) $(a, \alpha) + (b, \beta) := (a + b, \alpha + \beta)$,
- (ii) $\beta(a, \alpha) := (\beta a, \beta \alpha)$,
- (iii) $(a, \alpha)(b, \beta) := (ab + \alpha\beta + \beta a, \alpha\beta)$.

Then $\mathcal{A} \times \mathbb{F}$ is an algebra with identity $(0, 1)$.

Proof. test ■

Proposition 1.1.2. Let $\mathcal{A}$ be an algebra over $\mathbb{F}$ in which $(\mathcal{A}, \|\cdot\|_{\mathcal{A}})$ is a Banach space. For every $a \in \mathcal{A}$, suppose that the maps $x \mapsto ax$ and $x \mapsto xa$ of $\mathcal{A} \rightarrow \mathcal{A}$ are continuous. Let $\mathcal{A} \times \mathbb{F}$ be the algebra with operations given from Proposition ?? . For $a \in \mathcal{A}$, define $L_a : \mathcal{A} \times \mathbb{F} \rightarrow \mathcal{A} \times \mathbb{F}$ by $L_a(x, \xi) := (ax + \xi a, 0)$.

- (1) Define $\|(a, \alpha)\|_{\mathcal{A} \times \mathbb{F}} := \|a\|_{\mathcal{A}} + |\alpha|$. Then $\mathcal{A} \times \mathbb{F}$ is a Banach space and an algebra with identity $(0, 1)$.
- (2) The map $a \rightarrow (a, 0)$ is an isometric isomorphism between $\mathcal{A}$ and $\mathcal{A} \times \{0\}$.
- (3) For each $a \in \mathcal{A}$, we have $L_a \in \mathcal{B}(\mathcal{A} \times \mathbb{F})$.
- (4) Define $\|a\|_* := \|L_a\|_{\mathcal{B}(\mathcal{A} \times \mathbb{F})}$. Then $\|\cdot\|_*$ is equivalent to the norm $\|\cdot\|_{\mathcal{A}}$, and in particular $(\mathcal{A}, \|\cdot\|_*)$ is a Banach algebra.

Proof. ■

Parts (1) and (2) show that $\mathcal{A}$ may be identified isometrically with the ideal $\mathcal{A} \times \{0\}$ of the unital algebra $\mathcal{A} \times \mathbb{F}$, whose identity is $(0, 1)$. Thus, every algebra may be embedded isometrically into a unital algebra. When an algebra is already unital, it makes sense then to denote its identity by 1. We will use this convention throughout the rest of the paper.

Parts (3) and (4) show that the usual submultiplicativity condition on the norm need not be assumed. Indeed, if $\mathcal{A}$ is complete and multiplication is continuous, then the norm $\|a\|_*$ is equivalent to the original norm on $\mathcal{A}$ satisfying the submultiplicativity condition.

§ 1.2. Ideals and Quotients

Definition 1.2.1. Let $\mathcal{A}$ be an algebra. A *left ideal* (resp. *right ideal*) of $\mathcal{A}$ is a subalgebra $\mathcal{M}$ of $\mathcal{A}$ such that, for all $a \in \mathcal{A}$ and $x \in \mathcal{M}$, then $ax \in \mathcal{M}$ (resp. $xa \in \mathcal{M}$). If $\mathcal{M}$ is both a left and right ideal, then $\mathcal{M}$ is simply called an *ideal* (or *bilateral ideal*).

Definition 1.2.2. Let $\mathcal{A}$ be a unital algebra. An element $a \in \mathcal{A}$ is *left invertible* (resp. *right invertible*) if there exists an element $x \in \mathcal{A}$ such that $xa = 1$ (resp. $ax = 1$). If an element $a \in \mathcal{A}$ is both left and right invertible, then a is simply called *invertible*.

Proposition 1.2.1. Let $\mathcal{A}$ be a unital algebra.

- (1) If $a \in \mathcal{A}$ is invertible, there is a unique element a^{-1} such that $aa^{-1} = a^{-1}a = 1$.
- (2) If $a \in \mathcal{A}$ is left invertible, and $\mathcal{M}$ is a left ideal in $\mathcal{A}$ containing a , then $\mathcal{M} = \mathcal{A}$.

Proof. (1) Since a is invertible, there exists $x, y \in \mathcal{A}$ such that $xa = 1 = ay$. Hence $y = 1y = (xa)y = x(ay) = x1 = x$.

(2) Since $a \in \mathcal{A}$ is left invertible, there exists $x \in \mathcal{A}$ such that $xa = 1$. Since $\mathcal{M}$ is a left ideal and $a \in \mathcal{M}$, it must be the case that $1 = xa \in \mathcal{M}$. Since $1 \in \mathcal{M}$, we have $y = y1 \in \mathcal{M}$ for all $y \in \mathcal{A}$. Thus $\mathcal{A} \subseteq \mathcal{M}$. ■

Part (2) of Proposition ?? says that every element of a proper left ideal is necessarily not left invertible.

Lemma 1.2.2. If $\mathcal{A}$ is a unital Banach algebra and $x \in \mathcal{A}$ such that $\|x - 1\| < 1$, then x is invertible.

Proof. Define $y := 1 - x$. Note that:

$$\begin{aligned} \|y\| &= \|1 - x\| \\ &= \|x - 1\| \\ &< 1. \end{aligned}$$

Since $\mathcal{A}$ is a Banach algebra, it follows via induction that:

$$\|y^n\| \leq \|y\|^n.$$

Since $\|y\| < 1$, the series $\sum_{n=0}^{\infty} \|y\|^n$ converges, hence by the comparison test $\sum_{n=0}^{\infty} \|y^n\|$ converges. Because $\mathcal{A}$ is complete, $z := \sum_{n=0}^{\infty} y^n$ also converges. Taking $z_k := \sum_{n=0}^k y^n$, we can see:

$$\begin{aligned} z_k(1 - y) &= \sum_{n=0}^k y^n - \sum_{n=0}^k y^{n+1} \\ &= 1 - y^{k+1}. \end{aligned}$$

It is clear that $\sum_{n=0}^{\infty} y^{n+1}$ converges by the same reasoning from above. Hence $y^{n+1} \rightarrow 0$ as $n \rightarrow \infty$. This gives:

$$\begin{aligned} z(1 - y) &= \lim_{n \rightarrow \infty} z_n(1 - y) \\ &= \lim_{n \rightarrow \infty} 1 - y^{n+1} \\ &= 1. \end{aligned}$$

An identical argument will show that $(1 - y)z = 1$. Thus $(1 - y) = 1 - (1 - x) = x$ is invertible. ■

The following fact surfaced in the preceeding proof.

Corollary 1.2.3. *Let $\mathcal{A}$ be a unital Banach algebra. If $a \in \mathcal{A}$ and $\|a\| < 1$, then $(1 - a)^{-1} = \sum_{k=0}^{\infty} a^k$.*

Lemma 1.2.4. *Let $\mathcal{A}$ be a unital Banach algebra. If $b_0 a_0 = 1$ and $\|a - a_0\| < \|b_0\|^{-1}$, then a is left invertible.*

Proof. Rearranging $\|a - a_0\| < \|b_0\|^{-1}$ yields $\|b_0 a - 1\| < 1$. By Lemma 1.2.2, the element $b_0 a$ is invertible. Hence there exists $x \in \mathcal{A}$ such that $x b_0 a = 1$, implying that a is left invertible. ■

Theorem 1.2.5. *If $\mathcal{A}$ is a unital Banach algebra, then:*

- (i) $G_l := \{a \in \mathcal{A} : a \text{ is left invertible}\},$
- (ii) $G_r := \{a \in \mathcal{A} : a \text{ is right invertible}\},$
- (iii) $G := \{a \in \mathcal{A} : a \text{ is invertible}\},$

are all open subsets of $\mathcal{A}$. Moreover, the map $a \mapsto a^{-1}$ of $G \rightarrow G$ is continuous.

Proof. Let $a_0 \in G_l$. If $a \in B(a_0, \|b_0\|^{-1})$, then $\|a - a_0\| < \|b_0\|^{-1}$, so by Lemma 1.2.4 a is left invertible. Thus $a_0 \in B(a_0, \|b_0\|^{-1}) \subseteq G_l$, establishing G_l as open. It follows similarly that G_r is open, and since $G = G_l \cap G_r$, it is open as well.

To prove that $a \mapsto a^{-1}$ is continuous, we first prove an intermediary result. Let $\{a_n\}$ be a

sequence in G such that $a_n \rightarrow 1$. Let $0 < \delta < 1$ and suppose $\|a_n - 1\| < \delta$. By Corollary 1.2.3:

$$\begin{aligned} a_n^{-1} &= (1 - (1 - a_n))^{-1} \\ &= \sum_{k=0}^{\infty} (1 - a_n)^k \\ &= 1 + \sum_{k=1}^{\infty} (1 - a_n)^k. \end{aligned}$$

Hence $a_n^{-1} - 1 = \sum_{k=1}^{\infty} (1 - a_n)^k$. Thus:

$$\begin{aligned} \|a_n^{-1} - 1\| &= \left\| \sum_{k=1}^{\infty} (1 - a_n)^k \right\| \\ &\leq \sum_{k=1}^{\infty} \|1 - a_n\|^k \\ &= \sum_{k=1}^{\infty} \delta^k \\ &= \frac{\delta}{1 - \delta}. \end{aligned}$$

If $\epsilon > 0$ is given, then δ can be chosen such that $\frac{\delta}{1-\delta} < \epsilon$. So $\|a_n - 1\| < \delta$ implies $\|a_n^{-1} - 1\| < \epsilon$. Hence $a_n^{-1} \rightarrow 1$. With this, let $a \in G$ be arbitrary and suppose $\{a_n\}$ is a sequence in G such that $a_n \rightarrow a$. Then $a^{-1}a_n \rightarrow 1$. The intermediary result above gives $a_n^{-1}a = (a^{-1}a_n)^{-1} \rightarrow 1$. Hence $a_n^{-1} = a_n^{-1}aa^{-1} \rightarrow a^{-1}$. ■

Definition 1.2.3. A *maximal ideal* is a proper ideal that is contained in no larger proper ideal. The set of maximal ideals of a given algebra (or ring) is denoted by Spec_m .

Corollary 1.2.6. Let $\mathcal{A}$ be a unital Banach algebra.

- (1) The closure of a proper left (resp. right, bilateral) ideal is a proper left (resp. right, bilateral) ideal.
- (2) A maximal left (resp. right, bilateral) ideal is closed.

Proof. (1) Let $\mathcal{M}$ be a proper left ideal of $\mathcal{A}$. The proof showing $\text{cl}(\mathcal{M})$ is a subalgebra is omitted, but the idea can be replicated from the following. Let $a \in \mathcal{A}$ and $x \in \text{cl}(\mathcal{M})$. There exists a sequence $\{x_n\}$ in $\mathcal{M}$ such that $x_n \rightarrow x$. Then $\{ax_n\}$ is a sequence in $\mathcal{M}$ such that $ax_n \rightarrow ax$. Thus $ax \in \text{cl}(\mathcal{M})$, so $\text{cl}(\mathcal{M})$ is a left ideal. Let G_l denote the elements of $\mathcal{A}$ which are left invertible. By part (2) of Proposition 1.2.1, $\mathcal{M} \cap G_l = \emptyset$, which implies that $\mathcal{M} \subseteq \mathcal{A} \setminus G_l$. Since $\mathcal{A} \setminus G_l$ is closed, and since $\text{cl}(\mathcal{M})$ is the smallest closed set containing $\mathcal{M}$, we have $\text{cl}(\mathcal{M}) \subseteq \mathcal{A} \setminus G_l$. Since G_l is necessarily nonempty, $\text{cl}(\mathcal{M}) \neq \mathcal{A}$, hence it is proper.

(2) If $\mathcal{M}$ is a maximal left ideal, then $\text{cl}(\mathcal{M})$ is a proper ideal by (1). Hence $\mathcal{M} = \text{cl}(\mathcal{M})$ by maximality. ■

Theorem 1.2.7. If $\mathcal{A}$ is a Banach algebra and $\mathcal{M}$ is a proper closed ideal in $\mathcal{A}$, then $\mathcal{A}/\mathcal{M}$ is a Banach algebra with norm given by:

$$\|a + \mathcal{M}\| := \inf\{\|a + x\| : x \in \mathcal{M}\}.$$

If $\mathcal{A}$ has an identity, so does $\mathcal{A}/\mathcal{M}$.

Proof. The proof that $\mathcal{A}/\mathcal{M}$ is a normed linear space with well-defined operations given by $(a + \mathcal{M}) + (b + \mathcal{M}) = (a + b) + \mathcal{M}$ and $\alpha(a + \mathcal{M}) = (\alpha a) + \mathcal{M}$ is omitted. We will first show that $\mathcal{A}/\mathcal{M}$ is a Banach space. Let $\{a_n + \mathcal{M}\}_n$ be a Cauchy sequence in $\mathcal{A}/\mathcal{M}$. There is a subsequence $\{a_{n_k} + \mathcal{M}\}_k$ such that:¹

$$\|(a_{n_k} + \mathcal{M}) - (a_{n_{k+1}} + \mathcal{M})\| = \|(a_{n_k} - a_{n_{k+1}}) + \mathcal{M}\| < 2^{-k}.$$

Let $y_1 = 0$. By the definition of $\|\cdot\|$ on $\mathcal{A}/\mathcal{M}$, choose $y_2 \in \mathcal{M}$ such that:

$$\|a_{n_1} - a_{n_2} + y_2\| \leq \|a_{n_1} - a_{n_2} + \mathcal{M}\| + 2^{-1} < 2 \cdot 2^{-1}.$$

With y_2 fixed, choose $y_3 \in \mathcal{M}$ such that:

$$\begin{aligned} \|(a_{n_2} + y_2) - (a_{n_3} - y_2)\| &= \|a_{n_2} - a_{n_3} + (y_2 - y_3)\| \\ &\leq \|a_{n_2} - a_{n_3} + \mathcal{M}\| + 2^{-2} \\ &< 2 \cdot 2^{-2}. \end{aligned}$$

Inductively, we obtain a sequence $\{y_k\}_k$ in $\mathcal{M}$ such that:

$$\|(a_{n_k} + y_k) - (a_{n_{k+1}} + y_{k+1})\| < 2 \cdot 2^{-k}.$$

In particular, for $j < k$ we have:

$$\begin{aligned} \|(a_{n_k} + y_j) - (a_{n_k} + y_k)\| &\leq \|(a_{n_j} + y_j) - (a_{n_{j+1}} + y_{j+1})\| + \dots + \|(a_{n_{k-1}} + y_{k-1}) - (a_{n_k} + y_k)\| \\ &\leq \sum_{i=j}^k 2 \cdot 2^{-i} \\ &\leq \sum_{i=j}^{\infty} 2 \cdot 2^{-i} \\ &= 2^{2-j}. \end{aligned}$$

Thus $\{a_{n_k} + y_k\}_k$ is a Cauchy sequence in $\mathcal{A}$. Since $\mathcal{A}$ is complete, there exists $a_0 \in \mathcal{A}$ such that $a_{n_k} + y_k \rightarrow a_0$. Since the projection $\pi : \mathcal{A} \rightarrow \mathcal{A}/\mathcal{M}$ is continuous, we have:

$$\begin{aligned} \lim_{k \rightarrow \infty} a_{n_k} + \mathcal{M} &= \lim_{k \rightarrow \infty} \pi(a_{n_k} + y_k) \\ &= \pi \left(\lim_{k \rightarrow \infty} a_{n_k} + y_k \right) \\ &= \pi(a_0) \\ &= a_0 + \mathcal{M}. \end{aligned}$$

Since $\{a_n + \mathcal{M}\}_n$ is a Cauchy sequence with a convergent subsequence, it is convergent. Thus $\mathcal{A}/\mathcal{M}$ is a Banach space.

¹Suppose $\{x_n\}$ is a Cauchy sequence. For $\epsilon = 2^{-1}$, choose N_1 such that $m, n \geq N_1$ implies $\|x_m - x_n\| < 2^{-1}$. Now choose $n_1 \geq N_1$. For $\epsilon = 2^{-2}$, choose N_2 such that $m, n \geq N_2$ implies $\|x_m - x_n\| < 2^{-2}$. Choose $n_2 \geq \max\{N_2, n_1 + 1\}$. Since $n_1, n_2 \geq N_1$, we have $\|x_{n_2} - x_{n_1}\| < 2^{-1}$. Inductively, we obtain a subsequence $\{x_{n_k}\}_k$ satisfying the above property.

The proof that $\mathcal{A}/\mathcal{M}$ is an algebra with well-defined multiplication given by $(a + \mathcal{M})(b + \mathcal{M}) = ab + \mathcal{M}$ is omitted. Let $a, b \in \mathcal{A}$ and $u, v \in \mathcal{M}$. Then $(a + u)(b + v) = ab + (av + ub + uv) \in ab + \mathcal{M}$. Hence:

$$\begin{aligned} \|(a + \mathcal{M})(b + \mathcal{M})\| &= \|ab + \mathcal{M}\| \\ &\leq \|(a + u)(b + v)\|. \end{aligned}$$

Taking the infimum over all $u, v \in \mathcal{M}$ gives $\|(a + \mathcal{M})(b + \mathcal{M})\| \leq \|a + \mathcal{M}\| \|b + \mathcal{M}\|$. Thus $\mathcal{A}/\mathcal{M}$ is a Banach algebra. Moreover, $1 + \mathcal{M}$ is indeed the identity of $\mathcal{A}/\mathcal{M}$ as $(1 + \mathcal{M})(a + \mathcal{M}) = (1a + \mathcal{M}) = a + \mathcal{M} = (a1 + \mathcal{M}) = (a + \mathcal{M})(1 + \mathcal{M})$. ■

§ 1.3. The Spectrum

Definition 1.3.1. If $\mathcal{A}$ is a unital Banach algebra and $a \in \mathcal{A}$, the *spectrum* of a is:

$$\sigma(a) := \{a \in \mathbb{F} : a - \alpha \text{ is not invertible}\}.$$

The *left spectrum* of a is:

$$\sigma_l(a) := \{a \in \mathbb{F} : a - \alpha \text{ is not left invertible}\}.$$

The *right spectrum* of a , denoted $\sigma_r(a)$, is defined similarly.

Definition 1.3.2. If $\mathcal{A}$ is a unital Banach algebra and $a \in \mathcal{A}$, the *resolvent set* of a is $\rho(a) := \mathbb{F} \setminus \sigma(a)$. The *left* and *right resolvents* of a are $\rho_l(a) := \mathbb{F} \setminus \sigma_l(a)$ and $\rho_r(a) := \mathbb{F} \setminus \sigma_r(a)$.

Example 1.3.1. Let X be compact. If $f \in C(X)$, then $\sigma(f) = f(X)$. Indeed, let $\alpha \in f(X)$. Then $\alpha = f(x_0)$ for some $x_0 \in X$. Hence $0 = f(x_0) - \alpha \in (f - \alpha)(X)$. It must be the case that $(f - \alpha)^{-1}$ does not exist, so $\alpha \in \sigma(f)$. Conversely, if $f - \alpha$ is not invertible, then $(f - \alpha)^{-1}$ does not exist. This means $0 = f(x_0) - \alpha$ for some $x_0 \in X$, hence $\alpha = f(x_0)$.

Example 1.3.2. Let X be a Banach space. If $A \in \mathcal{B}(X)$, then $\sigma(A) = \{\alpha \in \mathbb{F} : \text{either } \ker(A - \alpha) \neq 0 \text{ or } \text{im}(A - \alpha) \neq X\}$. Indeed, if $A - \alpha$ is not invertible, then it is either not injective or not surjective. In fact, this means that $\rho(A) = \mathbb{F} \setminus \sigma(A) = \{\alpha \in \mathbb{F} : A - \alpha \text{ is bijective}\}$. If $\alpha \in \rho(A)$, then $\alpha \notin \sigma(A)$, hence $\ker(A - \alpha) = 0$ and $\text{im}(A - \alpha) = X$. Hence $A - \alpha$ is bijective. Conversely, suppose $A - \alpha$ is bijective. By the Open Mapping Theorem, $A - \alpha$ is an open map. For any open subset $U \subseteq X$, the set $((A - \alpha)^{-1})^{-1}(U) = (A - \alpha)(U)$ is open. Thus $(A - \alpha)^{-1} \in \mathcal{B}(X)$, hence $\alpha \in \rho(A)$.

§ 1.4. The Riesz Functional Calculus

Definition 1.4.1. Let $G \subseteq \mathbb{C}$ be open. A closed rectifiable curve γ is *positively oriented* if for every $a \in G \setminus \{\gamma\}$, $n(\gamma; a)$ is either 0 or 1. In this case the *inside* of γ is $\text{ins } \gamma := \{a \in \mathbb{C} \setminus \{\gamma\} : n(\gamma; a) = 1\}$. The *outside* of γ is $\text{out } \gamma := \{a \in \mathbb{C} \setminus \{\gamma\} : n(\gamma; a) = 0\}$.

Definition 1.4.2. A curve $\gamma : [0, 1] \rightarrow \mathbb{C}$ is *simple* if $\gamma(s) = \gamma(t)$ implies either:

- $s = t$,
- $s = 0$ and $t = 1$, or
- $s = 1$ and $t = 0$.

Definition 1.4.3. If $\Gamma = \{\gamma_1, \dots, \gamma_m\}$ is a collection of closed rectifiable curves, then Γ is *positively oriented* if:

- $\{\gamma_i\} \cap \{\gamma_j\} = \emptyset$,
- for all $a \in \mathbb{C} \setminus \bigcup_{j=1}^m \{\gamma_j\}$, $n(\Gamma; a) := \sum_{j=1}^m n(\gamma_j; a)$ is either 0 or 1, and
- each γ_j is a simple curve.

The *inside* of Γ is $\text{ins } \Gamma := \{a \in \mathbb{C} \setminus \bigcup_{j=1}^m \{\gamma_j\} : n(\Gamma; a) = 1\}$. The *outside* of Γ is $\text{out } \Gamma := \{a \in \mathbb{C} \setminus \bigcup_{j=1}^m \{\gamma_j\} : n(\Gamma; a) = 0\}$.

Proposition 1.4.1. If G is an open subset of $\mathbb{C}$ and K is a compact subset of G , then there is a positively oriented system of curves $\Gamma = \{\gamma_1, \dots, \gamma_m\}$ in $G \setminus K$ such that $K \subseteq \text{ins } \Gamma$ and $\mathbb{C} \setminus G \subseteq \text{out } \Gamma$. The curves $\gamma_1, \dots, \gamma_m$ can be found such that they are infinitely differentiable.

Proof. ■

The main use of this proposition will be for $K = \sigma(a)$.

Definition 1.4.4. Let $\mathcal{A}$ be a unital Banach algebra and let $a \in \mathcal{A}$. The set of functions that are analytic in a neighborhood of $\sigma(a)$ is denoted by $\text{Hol}(a)$.

Remark. $\text{Hol}(a)$ is not a Banach algebra. The domain of $f + g$ and fg is $\text{Dom}(f) \cap \text{Dom}(g)$, where $\text{Dom}(f)$ and $\text{Dom}(g)$ are the domains of f and g , respectively.

Theorem 1.4.2 (Riesz Functional Calculus). Let $\mathcal{A}$ be a unital Banach algebra and $a \in \mathcal{A}$. Define $\rho : \text{Hol}(a) \rightarrow \mathcal{A}$ by:

$$f \mapsto \frac{1}{2\pi i} \int_{\Gamma} f(z)(z - a)^{-1} dz,$$

where, using Proposition 1.4.1, Γ is a positively oriented collection of curves in an open set $G \subseteq \mathbb{C}$ such that $\sigma(a) \subseteq \text{ins } \Gamma \subseteq G$.

- (a) The map ρ is well-defined; i.e., if $G \subseteq \mathbb{C}$ is open such that $\sigma(a) \subseteq G$, $\Gamma = \{\gamma_1, \dots, \gamma_m\}$ and $\Lambda = \{\lambda_1, \dots, \lambda_k\}$ are two positively oriented collections of curves in G such that $\sigma(a) \subseteq \text{ins } \Gamma \subseteq G$ and $\sigma(a) \subseteq \text{ins } \Lambda \subseteq G$, then:

$$\frac{1}{2\pi i} \int_{\Gamma} f(z)(z - a)^{-1} dz = \frac{1}{2\pi i} \int_{\Lambda} f(z)(z - a)^{-1} dz.$$

- (b) The map ρ is an algebra homomorphism.
- (c) If $f(z) = z^k$ for all z , then $\rho(f) = a^k$.
- (d) If $\{f, f_1, f_2, \dots\}$ is a sequence of analytic functions on an open set G such that $\sigma(a) \subseteq G$ and $f_n \rightarrow f$ uniformly on compact subsets of G , then $\rho(f_n) \rightarrow \rho(f)$.
- (e) If $f(z) = \sum_{k=0}^{\infty} \alpha_k z^k$ has radius of convergence strictly greater than $r(a)$, then $f \in \text{Hol}(a)$ and $\rho(f) = \sum_{k=0}^{\infty} \alpha_k a^k$.
- (f) The map ρ is unique; i.e., suppose $\tau : \text{Hol}(a) \rightarrow \mathcal{A}$ is an algebra homomorphism satisfying:
- if $f(z) = 1$ for all z , then $\tau(f) = 1$,
 - if $f(z) = z$ for all z , then $\tau(f) = a$,

- if $\{f, f_1, f_2, \dots\}$ is a sequence of analytic functions on an open set G such that $\sigma(a) \subseteq G$ and $f_n \rightarrow f$ uniformly on compact subsets of G , then $\tau(f_n) \rightarrow \tau(f)$.

Then $\tau(f) = \rho(f)$ for every $f \in \text{Hol}(a)$.

Proof. (a)

(b) Let $f, g \in \text{Hol}(a)$ and let G be an open neighborhood of $\sigma(a)$ on which both f and g are analytic. Let Γ be a positively oriented system of closed curves in G such that $\sigma(a) \subseteq \text{ins } \Gamma$. Let Λ be a positively oriented system of closed curves in G such that $(\text{ins } \Gamma) \cup \{\Gamma\} = \text{cl}(\text{ins}(\Gamma)) \subseteq \text{ins } \Lambda$. Then:

$$\begin{aligned}
 \rho(f)\rho(g) &= \left(\frac{1}{2\pi i} \int_{\Gamma} f(z)(z-a)^{-1} dz \right) \left(\frac{1}{2\pi i} \int_{\Lambda} g(y)(y-a)^{-1} dy \right) \\
 &= -\frac{1}{4\pi^2} \int_{\Gamma} \int_{\Lambda} f(z)g(y)(z-a)^{-1}(y-a)^{-1} dy dz \\
 &= -\frac{1}{4\pi^2} \int_{\Gamma} \int_{\Lambda} f(z)g(y) \left(\frac{(z-a)^{-1} - (y-a)^{-1}}{y-z} \right) dy dz \\
 &= -\frac{1}{4\pi^2} \int_{\Gamma} \int_{\Lambda} \left(\frac{f(z)g(y)(z-a)^{-1}}{y-z} - \frac{f(z)g(y)(y-a)^{-1}}{y-z} \right) dy dz \\
 &= -\frac{1}{4\pi^2} \int_{\Gamma} \int_{\Lambda} \frac{f(z)g(y)(z-a)^{-1}}{y-z} dy dz + \frac{1}{4\pi^2} \int_{\Lambda} \int_{\Gamma} \frac{f(z)g(y)(y-a)^{-1}}{y-z} dz dy \\
 &= -\frac{1}{4\pi^2} \int_{\Gamma} f(z) \left(\int_{\Lambda} \frac{g(y)}{y-z} dy \right) (z-a)^{-1} dz \\
 &\quad + \frac{1}{4\pi^2} \int_{\Lambda} g(y) \left(\int_{\Gamma} \frac{f(z)}{y-z} dz \right) (y-a)^{-1} dy.
 \end{aligned}$$

For $y \in \{\Lambda\}$, then $y \in \text{out } \Lambda$ and hence $\int_{\Gamma} \frac{f(z)}{y-z} dz = 0$ by Cauchy's Theorem. For $z \in \{\Gamma\}$, then $z \in \text{ins } \Lambda$ and hence $\int_{\Lambda} \frac{g(y)}{y-z} dy = 2\pi i g(z)$ by Cauchy's Integral Formula. Thus:

$$\begin{aligned}
 \rho(f)\rho(g) &= -\frac{1}{2\pi^2} \int_{\Gamma} f(z) 2\pi i g(z) (z-a)^{-1} dz \\
 &= \frac{1}{2\pi i} \int_{\Gamma} (fg)(z) (z-a)^{-1} dz \\
 &= \rho(fg).
 \end{aligned}$$

Now if Π is a positively oriented system of closed curves in G such that $\sigma(a) \subseteq \text{ins } \Pi$, then:

$$\begin{aligned}
 \rho(\alpha f + \beta g) &= \frac{1}{2\pi i} \int_{\Pi} (\alpha f + \beta g)(z-a)^{-1} dz \\
 &= \frac{\alpha}{2\pi i} \int_{\Pi} f(z)(z-a)^{-1} dz + \frac{\beta}{2\pi i} \int_{\Pi} g(z)(z-a)^{-1} dz \\
 &= \alpha \rho(f) + \beta \rho(g).
 \end{aligned}$$

Thus ρ is an algebra homomorphism.

(b)

(c)

(d)

(e)

(f) We will first show that $\rho(f) = \tau(f)$ whenever f is rational, and then extend to all of $\text{Hol}(a)$ via Runge's Theorem. For $n \geq 1$, our hypothesis gives $\tau(z^n) = \tau(z)^n = a^n = \rho(z^n)$, hence $\tau(p) = \rho(p)$ whenever p is a polynomial. Let q be a polynomial such that q never vanishes on $\sigma(a)$. Then $q^{-1} \in \text{Hol}(a)$. Also, $1 = \tau(q \cdot a^{-1}) = \tau(q)\tau(q^{-1}) = \rho(q)\tau(q^{-1})$. Thus $\rho(q)$ is invertible and its inverse is $\tau(q^{-1})$. But, by the Riesz Functional Calculus, it must be the case that $\rho(q^{-1}) = \rho(q)^{-1}$. Hence $\tau(q^{-1}) = \rho(q^{-1})$. Now if $f = \frac{p}{q}$ where p and q are polynomials and q never vanishes on $\sigma(a)$, then the above results give $\tau(f) = \tau(pq^{-1}) = \tau(p)\tau(q^{-1}) = \rho(p)\rho(q^{-1}) = \rho(f)$. With this fact, let $f \in \text{Hol}(a)$ and suppose f is analytic on an open set G such that $\sigma(a) \subseteq G$. By Runge's Theorem there are rational functions $\{f_n\}$ whose poles lie outside of G such that $f_n \rightarrow f$ uniformly on compact subsets of G . Hence each f_n is analytic on G ; i.e., $\{f_n\} \subset \text{Hol}(a)$. By our hypothesis, $\tau(f_n) \rightarrow \tau(f)$. But we showed that $\tau(f_n) = \rho(f_n)$, and $\rho(f_n) \rightarrow \rho(f)$ by (d). Hence $\tau(f) = \rho(f)$. ■

Remark. Parts (c) and (e) show that the Riesz Functional Calculus can be interpreted as an "evaluation by a ," and part (f) asserts that this evaluation is unique. Thus, it makes sense to define the Riesz Functional Calculus $\rho(f)$ as $f(a)$. This convention will be used throughout the rest of the paper.

Proposition 1.4.3. *For all $f, g \in \text{Hol}(a)$, we have $f(a)g(a) = g(a)f(a)$.*

Proof. Note that $f(a)g(a) = \rho(fg) = \rho(gf) = g(a)f(a)$. ■

Proposition 1.4.4. *If $a, b \in \mathcal{A}$ such that $ab = ba$, and $f \in \text{Hol}(a)$, then $f(a)b = bf(a)$.*

Proof. Let p be a polynomial. Since $ab = ba$, we have $a^n b = b a^n$ for all $n \in \mathbb{N}$. By the distributive property, this gives $p(a)b = bp(a)$. Now let q be a polynomial such that its inverse never vanishes on $\sigma(a)$. Then $q(a)b = bq(a)$ as was just shown, but by multiplying $(q(a))^{-1}$ on both sides we see that $(q(a))^{-1}b = b(q(a))^{-1}$. By the Riesz Functional Calculus, $(q(a))^{-1} = q^{-1}(a)$. Hence if $r = \frac{p}{q}$ is a rational function which never vanishes on $\sigma(a)$, we have:

$$\begin{aligned} r(a)b &= (pq^{-1})(a)b \\ &= p(a)q^{-1}(a)b \\ &= bp(a)q^{-1}(a) \\ &= br(a). \end{aligned}$$

Using Runge's Theorem, if $f \in \text{Hol}(a)$ such that f is analytic on an open set G containing $\sigma(a)$, there is a sequence $\{f_n\}$ of rational functions which converge to f on compact subsets of G . Since $f_n(a)b = bf_n(a)$ for each n , we have:

$$\begin{aligned} f(a)b &= \lim_{n \rightarrow \infty} f_n(a)b \\ &= \lim_{n \rightarrow \infty} bf_n(a) \\ &= bf(a) \end{aligned} \quad \blacksquare$$

Theorem 1.4.5 (The Spectral Mapping Theorem). *If $a \in \mathcal{A}$ and $f \in \text{Hol}(a)$, then $\sigma(f(a)) = f(\sigma(a))$.*

Proof. Let $f(\alpha) \in \sigma(f(a))$. Then $\alpha \in \sigma(a)$. Towards contradiction, suppose $f(\alpha) \notin \sigma(f(a))$. Then $f(a) - f(\alpha)$ is invertible for some $z \in \mathbb{C}$. Consider a function g defined by $f(z) - f(\alpha) = (z - \alpha)g(z)$. Then g is analytic on a neighborhood of $\sigma(a)$; i.e., $g \in \text{Hol}(a)$. The Riesz Functional Calculus of g is $f(a) - f(\alpha) = (a - \alpha)g(a)$. Hence $(a - \alpha)$ is invertible with inverse $g(a)(f(a) - f(\alpha))^{-1}$. This contradicts $\alpha \in \sigma(a)$, concluding that $f(\alpha) \in \sigma(f(a))$. Thus $f(\sigma(a)) \subseteq \sigma(f(a))$.

Conversely, suppose $\beta \notin f(\sigma(a))$. Then $\beta \neq f(z)$ for every $z \in \sigma(a)$. Hence $f(z) - \beta$ is nonzero on $\sigma(a)$. Thus $g(z) := (f(z) - \beta)^{-1}$ is in $\text{Hol}(a)$. After applying the Riesz Functional Calculus, we get $g(a)(f(a) - \beta)^{-1} = 1$. Hence $\beta \notin \sigma(f(a))$, establishing $\sigma(f(a)) \subseteq f(\sigma(a))$. ■

Proposition 1.4.6. *application*

§ 1.5. Abelian Banach Algebras

Throughout, assume that all Banach algebras contain an identity.

Theorem 1.5.1 (Gelfand-Mazur). *If $\mathcal{A}$ is a Banach algebra that is also a division ring, then $\mathcal{A}$ is isometrically isomorphic to $\mathbb{C}$.*

Proof. Define $\varphi : \mathbb{C} \rightarrow \mathcal{A}$ by $z \mapsto z1_{\mathcal{A}}$. This is clearly an algebra homomorphism, and moreover:

$$\begin{aligned} \|\varphi(z)\| &= |z| \|1_{\mathcal{A}}\| \\ &= |z|. \end{aligned}$$

Hence φ is an isometry (and thus an injection). It remains to show surjectivity. If $a \in \mathcal{A}$, then $\sigma(a) \neq \emptyset$. If $z \in \sigma(a)$, then $a - z$ is not invertible. But $\mathcal{A}$ is a division ring; i.e., $\mathcal{A}^\times = \mathcal{A} \setminus \{0\}$. Thus $a - z = 0$, hence $a = z = z1_{\mathcal{A}} = \varphi(z)$. ■

Corollary 1.5.2. *Let $\mathcal{A}$ be an abelian Banach algebra and $\mathcal{M}$ a proper ideal. Then $\mathcal{M}$ is maximal if and only if $\mathcal{A}/\mathcal{M}$ is isometrically isomorphic to $\mathbb{C}$.*

Proof. The ideal $\mathcal{M}$ is maximal if and only if there are no ideals I with $\mathcal{M} \subset I \subset \mathcal{A}$. The ideals of $\mathcal{A}$ containing $\mathcal{M}$ correspond bijectively with the ideals of $\mathcal{A}/\mathcal{M}$, so $\mathcal{M}$ is maximal if and only if the ideals of $\mathcal{A}/\mathcal{M}$ are 0 and $\mathcal{A}/\mathcal{M}$. Equivalently, $\mathcal{M}$ is maximal if and only if $\mathcal{A}/\mathcal{M}$ is a field. By Theorem 1.2.7 and the Gelfand-Mazur Theorem, $\mathcal{M}$ is maximal if and only if $\mathcal{A}/\mathcal{M} \cong \mathbb{C}$. ■

Definition 1.5.1. A *character* of an abelian Banach algebra $\mathcal{A}$ is a nonzero homomorphism $\chi : \mathcal{A} \rightarrow \mathbb{C}$. The set of all characters of $\mathcal{A}$ is denoted by $\Sigma_{\mathcal{A}}$.

Lemma 1.5.3. *Every $\chi \in \Sigma_{\mathcal{A}}$ is surjective.*

Proof. Let $z \in \mathbb{C}$. For any $a \in \mathcal{A}$ we have $\chi\left(\frac{za}{\chi(a)}\right) = z \frac{\chi(a)}{\chi(a)} = z$. ■

Proposition 1.5.4. *Let $\mathcal{A}$ be an abelian Banach algebra and $\mathcal{M}$ a maximal ideal of $\mathcal{A}$. Then $\mathcal{M} = \ker \chi$ for some $\chi \in \Sigma_{\mathcal{A}}$.*

Proof. Let $\pi : \mathcal{A} \rightarrow \mathcal{A}/\mathcal{M}$ be the natural projection. By Corollary 1.5.2 and the Gelfand-Mazur Theorem, $\varphi : \mathcal{A}/\mathcal{M} \rightarrow \mathbb{C}$ given by $z1_{\mathcal{A}} + \mathcal{M} \mapsto z$ is an isomorphism. Define $\chi : \mathcal{A} \rightarrow \mathbb{C}$ by $\chi := \varphi \circ \pi$. Then $\chi \in \Sigma_{\mathcal{A}}$ and:

$$\begin{aligned} \ker \chi &= \{a \in \mathcal{A} : \chi(a) = 0\} \\ &= \{a \in \mathcal{A} : \varphi(a + \mathcal{M}) = 0\} \\ &= \mathcal{M}. \end{aligned}$$

■

Proposition 1.5.5. *If $\chi \in \Sigma_{\mathcal{A}}$, then $\ker \chi$ is a maximal ideal.*

Proof. By the First Isomorphism Theorem, $\mathcal{A}/\ker \chi \cong \chi(\mathcal{A})$. Lemma 1.5.3 says that $\chi(\mathcal{A}) = \mathbb{C}$, hence by Corollary 1.5.2 $\ker \chi$ is maximal. ■

Theorem 1.5.6. *Let $\mathcal{A}$ be an abelian Banach algebra. There is a bijection between $\Sigma_{\mathcal{A}}$ and $\text{Spec}_{\mathfrak{m}}(\mathcal{A})$.*

Proof. Define $f : \Sigma_{\mathcal{A}} \rightarrow \text{Spec}_{\mathfrak{m}}(\mathcal{A})$ by $\chi \mapsto \ker \chi$. Suppose $\chi_1, \chi_2 \in \Sigma_{\mathcal{A}}$ with $f(\chi_1) = f(\chi_2)$. Then $\ker \chi_1 = \ker \chi_2$. Note that for any $a \in \mathcal{A}$, if $\chi(a) = 0$ then $a \in \ker \chi_1 = \ker \chi_2$, hence $\chi_2(a) = 0$. By Corollary A.5.3, $\chi_1 = \alpha \chi_2$. But notice $\alpha = \alpha \chi_2(1) = \chi_1(1) = 1$, hence $\chi_1 = \chi_2$. Thus f is injective. Proposition 1.5.4 shows that f is surjective, hence $\Sigma_{\mathcal{A}}$ and $\text{Spec}_{\mathfrak{m}}(\mathcal{A})$ have a bijective correspondence. ■

Corollary 1.5.7. *If $\mathcal{A}$ is an abelian Banach algebra, then $\Sigma_{\mathcal{A}} \subseteq \mathcal{A}^*$; i.e., every character of $\mathcal{A}$ is continuous.*

Proof. Let $\chi \in \Sigma_{\mathcal{A}}$. Then $\ker \chi$ is a maximal ideal, hence it is closed. If it were the case that χ is not continuous, then it is unbounded. Hence there exists a sequence $\{a_n\}_n \subseteq \mathcal{A}$ such that $\chi(a_n) \rightarrow \infty$ and $n \rightarrow \infty$. If $a \notin \ker \chi$, consider the sequence $\{a'_n\}_n$ given by:

$$a'_n := a - \frac{\chi(a)}{\chi(a_n)} a_n.$$

Then $\chi(a'_n) = 0$ for all n , hence $\{a'_n\}_n \subseteq \ker \chi$. But $a'_n \rightarrow a \notin \ker \chi$, so $\ker \chi$ is not closed. Since we've proven the contrapositive, every $\chi \in \Sigma_{\mathcal{A}}$ is continuous; i.e., $\Sigma_{\mathcal{A}} \subseteq \mathcal{A}^*$. ■

Corollary 1.5.8. *If $\mathcal{A}$ is an abelian Banach algebra and $\chi \in \Sigma_{\mathcal{A}}$, then $\|\chi\| = 1$.*

Proof. ■

Definition 1.5.2.

Chapter A

Topological Vector Spaces and Weak Topologies

Throughout, unless otherwise stated, assume $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

§ A.1. Preliminaries

Throughout, assume X to be a topological space.

Definition A.1.1. If $E \subset X$, a *neighborhood* of E is a set $A \subset X$ such that $E \subset \text{int } A$.

Definition A.1.2. If $\mathcal{E} \subset \mathcal{P}(X)$, the topology *generated* by $\mathcal{E}$, denoted by $\mathcal{T}(\mathcal{E})$, is defined as the intersection of all topologies on X containing $\mathcal{E}$. The generating set $\mathcal{E}$ is called a *subbase*.

Definition A.1.3. If $\mathcal{T}$ is a topology on X , a *local base* for $\mathcal{T}$ at $x \in X$ is a family $\mathcal{N} \subseteq \mathcal{T}$ such that:

- $x \in V$ for all $V \in \mathcal{N}$;
- if $U \in \mathcal{T}$ and $x \in U$, there exists $V \in \mathcal{N}$ such that $x \in V$ and $V \subseteq U$.

Definition A.1.4. A *base* for a topology $\mathcal{T}$ on X is a family $\mathcal{B} \subseteq \mathcal{T}$ that contains a local base for $\mathcal{T}$ at every $x \in X$.

Example A.1.1. If X is a metric space, all open balls centered at $x \in X$ form a local base. The collection of all open balls in X form a base.

Proposition A.1.1. If $\mathcal{T}$ is a topology on X and $\mathcal{E} \subset \mathcal{T}$, then $\mathcal{E}$ is a base if and only if every nonempty $U \in \mathcal{T}$ is a union of members of $\mathcal{E}$.

Proof. ■

Proposition A.1.2. If $\mathcal{E} \subset \mathcal{P}(X)$, then $\mathcal{E}$ is a base for a topology on X if and only if:

- each $x \in X$ is contained in some $V \in \mathcal{E}$;
- If $U, V \in \mathcal{E}$ and $x \in U \cap V$, there exists $W \in \mathcal{E}$ such that $x \in W \subseteq (U \cap V)$.

Proof. ■

Proposition A.1.3. If $\mathcal{E} \subset \mathcal{P}(X)$, the topology $\mathcal{T}(\mathcal{E})$ consists of $\emptyset$, X , and all unions of finite intersections of members of $\mathcal{E}$.

Proof. ■

Proposition A.1.4. *Let $\mathcal{E} \subset \mathcal{P}(X)$. The collection of all finite intersections of elements in $\mathcal{E}$ form a base for $\mathcal{T}(\mathcal{E})$.*

Proof. ■

§ A.2. Vector Topologies and Separation Properties

Definition A.2.1. Suppose $\mathcal{T}$ is a topology on an $\mathbb{F}$ -vector space X such that:

- for every $x \in X$, the set $\{x\}$ is closed;
- the vector space operations are continuous with respect to $\mathcal{T}$.

Under these conditions, $\mathcal{T}$ is called a vector topology and X is said to be a topological vector space (abbrev. TVS).

Definition A.2.2. Let X be a TVS.

- (a) A set $C \subset X$ is said to be convex if $tC + (1 - t)C \subset C$ for every $0 \leq t \leq 1$.
- (b) A set $B \subset X$ is said to be balanced if $\alpha B \subset B$ for every $\alpha \in \mathbb{F}$ with $|\alpha| \leq 1$.

Proposition A.2.1. *Let X be a TVS.*

- (a) *The translation operator $T_a : X \rightarrow X$ given by $T_a(x) = a + x$ is a homeomorphism.*
- (b) *The multiplication operator $M_\lambda : X \rightarrow X$ given by $M_\lambda(x) = \lambda x$ is a homeomorphism.*

Proof. The continuity and bijectivity of T_a and M_λ follow from the vector space axioms. Their inverses, T_{-a} and $M_{1/\lambda}$, are also continuous due to the vector space axioms. ■

Remark. A consequence of Proposition A.2.1 is that every vector topology is *translation-invariant*; i.e., a set $E \subset X$ is open if and only if each of its translates $a + E$ is open. As such, any vector topology is completely determined by any of its local bases. The following proposition illustrates this.

Proposition A.2.2. *Let $\mathcal{T}$ be a vector topology on X . If $\mathcal{B}$ is a local base at 0, then every open set of X can be written as a union of translates of members of $\mathcal{B}$.*

Proof. ■

Definition A.2.3. If X is a TVS, then a (*canonical*) *local base* is a local base at 0

Lemma A.2.3. *Let X be a TVS. If W is a neighborhood of 0 in X , then there is a neighborhood U of 0 such that $U = -U$ and $U + U \subset W$.*

Proof. Let $P : X \times X \rightarrow X$ denote addition. Since addition is continuous, it is continuous at 0. Hence $P^{-1}(W)$ is a neighborhood of $(0, 0)$. In particular, there exists open sets V_1 and V_2 of X such that $(0, 0) \in V_1 \times V_2 \subseteq P^{-1}(W)$. Now define $U := V_1 \cap V_2 \cap (-V_1) \cap (-V_2)$. This is an open set of X containing zero which is symmetric and contained in both V_1 and V_2 . Thus:

$$U + U = P(U \times U) \subseteq P(V_1 \times V_2) \subseteq W. \quad \blacksquare$$

Lemma A.2.4. *Let X be a TVS. Suppose K and C are subsets of X where K is compact, C is closed, and $K \cap C = \emptyset$. Then 0 has a neighborhood V such that:*

$$(K + V) \cap (C + V) = \emptyset.$$

Proof. Suppose $x \in K$. Then $x \notin C$, implying that C^c is a neighborhood of x . By Proposition A.2.1, the translation $C^c - x$ is a neighborhood of 0, so by Lemma A.2.3 we can find a symmetric neighborhood V_x of 0 such that:

$$V_x + V_x + V_x \subset C^c - x.$$

This can be rewritten as:

$$x + V_x + V_x + V_x \subset C^c.$$

Claim: $(x + V_x + V_x) \cap (V_x + C) = \emptyset$. Indeed, if $z \in (x + V_x + V_x) \cap (V_x + C)$, then $z = x + v_1 + v_2$ and $z = v_3 + c$ for $v_1, v_2, v_3 \in V_x$ and $c \in C$. Equating the two yields $x + v_1 + v_2 - v_3 = c$, which is a contradiction as we showed $(x + V_x + V_x + V_x) \cap C = \emptyset$.

Since $x + V_x$ is a neighborhood of x , we can cover K as follows:

$$K \subset \bigcup_{x \in K} (x + V_x).$$

Since K is compact, there are finitely many points $x_1, \dots, x_n$ in K such that:

$$K \subset (x_1 + V_{x_1}) \cup \dots \cup (x_n + V_{x_n}).$$

Define $V := V_{x_1} \cap \dots \cap V_{x_n}$. Then:

$$K + V \subset \bigcup_{i=1}^n (x_i + V_{x_i} + V) \subset \bigcup_{i=1}^n (x_i + V_{x_i} + V_{x_i}).$$

By our claim, no term in the last union intersects $C + V$, establishing the proof. $\blacksquare$

Theorem A.2.5. *Let X be a TVS. Suppose K and C are subsets of X where K is compact, C is closed, and $K \cap C = \emptyset$. There exists disjoint open sets that contain K and C , respectively.*

Proof. From Lemma A.2.4, note that $K + V$ is a union of translates $x + V$ of V . Thus $K + V$ is an open set that contains K . Similarly, as C is the union of translates $x + C$, it is an open set containing C . $\blacksquare$

Corollary A.2.6. *If X is a TVS, and $\mathcal{B}$ is a local base for X , then every member of $\mathcal{B}$ contains the closure of some member of $\mathcal{B}$.*

Proof. Let $B \in \mathcal{B}$. Since $\{0\}$ is compact and $X \setminus B$ is closed, Lemma A.2.4 says there exists a neighborhood V of 0 such that:

$$(\{0\} + V) \cap (X \setminus B) = \emptyset.$$

This simplifies to:

$$V \cap (X \setminus B) = \emptyset.$$

It is a general fact that an open set which is disjoint from a set is also disjoint from its closure. Hence:

$$\overline{V} \cap (X \setminus B) = \emptyset.$$

This implies:

$$\overline{V} \subset B.$$

Since V is a neighborhood of 0 and $\mathcal{B}$ is a local base, we can find a $B' \in \mathcal{B}$ such that $B' \subset V$. Hence $\overline{B'} \subset \overline{V} \subset B$, establishing the claim. ■

Corollary A.2.7. *Every TVS is a Hausdorff space.*

Proof. Apply Lemma A.2.4 to a pair of distinct points in the place of K and C . ■

§ A.3. Local Convexity

Definition A.3.1. Let X be a TVS and $\mathcal{T}$ its vector topology.

- (a) X is locally convex if there is a local base $\mathcal{B}$ whose members are convex.
- (b) X is a Fréchet space if X is locally convex and its topology $\mathcal{T}$ is induced by a complete invariant metric.

Definition A.3.2. A seminorm on an $\mathbb{F}$ -vector space is a function $p : X \rightarrow \mathbb{R}$ such that:

- $p(x + y) \leq p(x) + p(y)$;
- $p(\alpha x) = |\alpha|p(x)$

for all $x, y \in X$ and $\alpha \in \mathbb{F}$.

Proposition A.3.1. Suppose p is a seminorm on an $\mathbb{F}$ -vector space X .

- (a) $p(0) = 0$.
- (b) $|p(x) - p(y)| \leq p(x - y)$.
- (c) $p(x) \geq 0$.
- (d) $\{x : p(x) = 0\}$ is a subspace of X .

Proof. ■

Definition A.3.3. A family $\mathcal{P}$ of seminorms on an $\mathbb{F}$ -vector space X is called separating if each $x \neq 0$ corresponds to at least one $p \in \mathcal{P}$ with $p(x) \neq 0$.

Definition A.3.4. Let X be a $\mathbb{F}$ -vector space.

- (a) A convex set $A \subset X$ is absorbing if for every $x \in X$, there exists a scalar $t > 0$ (possibly depending on x) such that $x \in tA$.
- (b) If A is an absorbing set, the Minkowski functional of A is a function $\mu_A : X \rightarrow \mathbb{F}$ defined by:

$$\mu_A(x) := \inf\{t : t^{-1}x \in A\}$$

Proposition A.3.2. Suppose p is a seminorm on an $\mathbb{F}$ -vector space X . The set $B := \{x : p(x) < 1\}$ is convex, balanced, absorbing, and $p = \mu_B$.

Proof. For any $x \in X$ and $\alpha \in \mathbb{F}$ we have:

$$\begin{aligned} p(\alpha x) &= |\alpha|p(x) \\ &\leq p(x) \\ &< 1. \end{aligned}$$

Thus B is balanced. For any $x, y \in X$ and $0 < t < 1$ we have:

$$\begin{aligned} p(tx + (1-t)y) &\leq p(tx) + p((1-t)y) \\ &= tp(x) + (1-t)p(y) \\ &\leq t + (1-t) \\ &= 1. \end{aligned}$$

Thus B is convex. If $x \in X$ and $s > p(x)$, we have:

$$\begin{aligned} p(s^{-1}x) &= s^{-1}p(x) \\ &< 1. \end{aligned}$$

Thus $s^{-1}x \in B$, hence B is absorbing. We will now show that $p = \mu_B$. Let $x \in X$ and $s \in \mathbb{F}$ such that $p(x) < s$. Then $p(s^{-1}x) < 1$, implying that $s \in \{t : t^{-1}x \in B\}$. This means $\mu_B(x) \leq s$, and taking the infimum over all s such that $p(x) < s$ gives $\mu_B(x) \leq p(x)$. Conversely, let $x \in X$ and $r \in \mathbb{F}$ such that $0 < r \leq p(x)$. Then $1 \leq p(r^{-1}x)$, implying that $r \notin \{t : t^{-1}x \in B\}$. We've established the inclusion $\{t : t^{-1}x \in B\} \subseteq (p(x), \infty)$, and taking the infimum of both sides gives $p(x) \leq \mu_B(x)$. Thus $p = \mu_B$. ■

Theorem A.3.3. Suppose A is a convex absorbing set in a vector space X .

- (a) $\mu_A(x + y) \leq \mu_A(x) + \mu_A(y)$.
- (b) $\mu_A(tx) = t\mu_A(x)$ if $t \geq 0$.
- (c) μ_A is a seminorm if A is balanced.
- (d) If $B = \{x : \mu_A(x) < 1\}$ and $C = \{x : \mu_A(x) \leq 1\}$, then $B \subset A \subset C$ and $\mu_B = \mu_A = \mu_C$.

Proof. ■

Theorem A.3.4. Let X be a TVS. Suppose $\mathcal{B}$ is a convex balanced local base in X .

- (a) For every $V \in \mathcal{B}$, we have $V = \{x \in X : \mu_V(x) < 1\}$.
- (b) $\{\mu_V : V \in \mathcal{B}\}$ is a separating family of continuous seminorms.

Proof. ■

Theorem A.3.5. Suppose $\mathcal{P}$ is a separating family of seminorms on a vector space X . Associate to each $p \in \mathcal{P}$ and to each $n \in \mathbb{N}$ the set:

$$V(p, n) := \left\{ x : p(x) < \frac{1}{n} \right\}.$$

Let $\mathcal{B}$ be the collection of all finite intersections of the sets $V(p, n)$, and let $\mathcal{B}'$ be the subset of $\mathcal{B}$ only containing neighborhoods of 0. Then $\mathcal{B}'$ is a convex balanced local base for a topology $\mathcal{T}$ on X such that:

- every $p \in \mathcal{P}$ is continuous;
- a set $E \subset X$ is bounded if and only if every $p \in \mathcal{P}$ is bounded on E .

Proof. ■

Theorem A.3.6. Let X be a TVS. Suppose:

- $\mathcal{B}$ is a convex balanced local base in X ;
- $\mathcal{P}$ is a separating family of seminorms on X .

Let $\mathcal{T}_1$ be the topology associated with $\mathcal{B}$. Let $\mathcal{T}_2$ be the topology associated to the convex balanced local base constructed using $\mathcal{P}$ as per Theorem A.3.5. Then $\mathcal{T}_1 = \mathcal{T}_2$.

Proof. ■

§ A.4. Topologies Induced by a Family of Sets

Definition A.4.1. Let $\mathcal{T}_1$ and $\mathcal{T}_2$ be two topologies on a set X . If $\mathcal{T}_1 \subset \mathcal{T}_2$ (that is, every set open in $\mathcal{T}_1$ is open in $\mathcal{T}_2$), then we say $\mathcal{T}_1$ is weaker than $\mathcal{T}_2$, or that $\mathcal{T}_2$ is stronger than $\mathcal{T}_1$.

Definition A.4.2. Let X be a set and $\{f_\alpha : X \rightarrow Y_\alpha\}_{\alpha \in I}$ be a collection of maps into some topological spaces Y_α . The topology generated by $\{f_\alpha^{-1}(U_\alpha) : \alpha \in I, U_\alpha \text{ open in } Y_\alpha\}$ is called the *initial topology* on X generated by $\{f_\alpha\}_{\alpha \in I}$.

Proposition A.4.1. The initial topology generated by $\{f_\alpha\}_{\alpha \in I}$ is the weakest topology making all the f_α continuous.

Proof. ■

Example A.4.1. If $\{X_\alpha\}_{\alpha \in I}$ is a family of topological spaces, the product topology on $\prod_{\alpha \in I} X_\alpha$ is the initial topology generated by $\{\pi_\alpha\}_{\alpha \in I}$, where each π_j is the projection onto X_j .

Definition A.4.3. A family $\mathcal{F}$ of maps on a set X is called *separating* if for any $x, y \in X$ with $x \neq y$, there exists a function $f \in \mathcal{F}$ such that $f(x) \neq f(y)$.

Proposition A.4.2. If $\{f_\alpha : X \rightarrow Y\}_\alpha$ is a separating family of maps, where X is a set and each Y_α is a Hausdorff space, then the initial topology on X generated by $\{f_\alpha\}_\alpha$ is Hausdorff.

Proof. ■

§ A.5. Dual Systems and Weak Topologies

Definition A.5.1. Let X be a TVS. The *algebraic dual* of X , denoted by $X^\#$, is the set of all linear functionals on X . The *continuous* (or *topological*) *dual* of X , denoted by X^* , is the subspace of $X^\#$ consisting of all continuous linear functions on X .

Definition A.5.2. Let X and Y be $\mathbb{F}$ -vector spaces and $b : X \times Y \rightarrow \mathbb{F}$ a bilinear form satisfying:

- $b(x_0, y) = 0$ for all $y \in Y$ implies $x_0 = 0$;
- $b(x, y_0) = 0$ for all $x \in X$ implies $y_0 = 0$.

The triple (X, Y, b) is called a *dual system*. We say that the function b places X and Y in *duality*. The function b is called the *canonical bilinear form* of duality, and is usually denoted by $\langle \cdot, \cdot \rangle$. The triple $(X, Y, \langle \cdot, \cdot \rangle)$ is more conveniently denoted by $\langle X, Y \rangle$.

Remark. If $\langle X, Y \rangle$ is a system, then by the symmetry of the above definition $\langle Y, X \rangle$ is also a system. Hence any proposition being made about $\langle X, Y \rangle$ also holds for $\langle Y, X \rangle$.

Example A.5.1. If X is an $\mathbb{F}$ -vector space, the map $\langle x, f \rangle \mapsto f(x)$ of $X \times X^\# \rightarrow \mathbb{F}$ is a bilinear form, and hence defines a dual system $\langle X, X^\# \rangle$. More can be said if X is a LCS: since X^* is a subspace of $X^\#$ which separates points in X , the restriction of $X \times X^\#$ onto $X \times X^*$ induces a dual system $\langle X, X^* \rangle$.

Proposition A.5.1. If $\langle X, Y \rangle$ is a dual system, then Y is isomorphic to a subspace of $X^\#$.

Proof. For every $y \in Y$, note that $\langle \cdot, y \rangle : X \rightarrow \mathbb{F}$ is a linear functional on X . Define $\varphi : Y \rightarrow X^\#$ by $\varphi(y) := \langle \cdot, y \rangle$. Since $\langle \cdot, \cdot \rangle$ is bilinear, φ is linear. Moreover, its kernel is clearly trivial, hence by the First Isomorphism Theorem $Y \cong \varphi(Y) \subseteq X^\#$. ■

Proposition A.5.2. Let $\langle X, Y \rangle$ be a dual system and suppose $\{y_1, \dots, y_n\}$ is a linearly independent subset of Y . There exists a linearly independent subset $\{x_1, \dots, x_n\}$ of X such that $\langle x_i, y_j \rangle = \delta_{ij}$.

Proof. ■

Corollary A.5.3. Let X be an $\mathbb{F}$ -vector space and let $g, f_1, f_2, \dots, f_n$ be linear functionals on X . For any $x \in X$, if we have $f_i(x) = 0$ for all $i = 1, \dots, n$ implies $g(x) = 0$, then g is a linear combination of $f_1, f_2, \dots, f_n$.

Proof. ■

Definition A.5.3. Let $\langle X, Y \rangle$ be a dual system. The *weak topology* on X , denoted $\sigma(X, Y)$, is the initial topology on X generated by $\{\langle \cdot, y \rangle : y \in Y\}$.

Proposition A.5.4. If $\langle X, Y \rangle$ is a dual system, then $(X, \sigma(X, Y))$ is a LCS.

Proof. Consider the collection of maps $\mathcal{P} := \{|\langle \cdot, y \rangle| : y \in Y\}$. By the bilinearity of $\langle \cdot, \cdot \rangle$ and properties of the absolute value, we have $|\langle x_1 + x_2, y \rangle| \leq |\langle x_1, y \rangle| + |\langle x_2, y \rangle|$ and $|\langle \alpha x, y \rangle| = |\alpha| |\langle x, y \rangle|$. Moreover, the contrapositive of the definition of $\langle \cdot, \cdot \rangle$ says that if $x \neq 0$, then there exists a $y \in Y$ such that $\langle x, y \rangle \neq 0$. Thus $\mathcal{P}$ is a separating family of seminorms, so by Theorem A.3.5 X is a LCS. ■

Theorem A.5.5. Let $\langle X, Y \rangle$ be a dual system. Then $(X, \sigma(X, Y))^* \cong Y$.

Proof. Define $\varphi : Y \rightarrow (X, \sigma(X, Y))^*$ by $\varphi(y)(x) := \langle x, y \rangle$. This map is clearly linear and its kernel is trivial. It remains to show surjectivity. Let $f \in (X, \sigma(X, Y))^*$. We showed in Proposition A.5.4 that $\{|\langle \cdot, y \rangle| : y \in Y\}$ is a family of seminorms generating the topology on $(X, \sigma(X, Y))$, hence by Proposition ?? there exists $y_1, \dots, y_n \in Y$ such that $|f(x)| \leq \gamma \max\{|\langle x, y_i \rangle| : i = 1, \dots, n\}$ for every $x \in X$ and for some $\gamma > 0$. If $\langle x, y_i \rangle = 0$ for every $i = 1, \dots, n$, then it must be the case that $f(x) = 0$. Hence by Corollary A.5.3 there exists $\alpha_1, \dots, \alpha_n \in \mathbb{F}$ such that $f = \sum_{i=1}^n \alpha_i \langle \cdot, y_i \rangle$. Note that $\sum_{i=1}^n \alpha_i y_i \in Y$, which gives:

$$\begin{aligned} \varphi\left(\sum_{i=1}^n \alpha_i y_i\right) &= \sum_{i=1}^n \alpha_i \varphi(y_i) \\ &= \sum_{i=1}^n \alpha_i \langle \cdot, y_i \rangle \\ &= f. \end{aligned}$$

Thus $(X, \sigma(X, Y))^* \cong Y$. ■

§ A.6. Weak* Topologies and the Banach-Alaoglu Theorem

Remark. If X is a TVS, then $\sigma(X, X^*)$ is a second topology on X . However, X^* has no topology to begin with. One might consider endowing X^* with $\sigma(X^*, X^{**})$, but there is a more natural choice. Recall that a dual system $\langle X, Y \rangle$ is symmetric; i.e., any property on $\langle X, Y \rangle$ also holds for $\langle Y, X \rangle$. Hence the dual system $\langle X, X^* \rangle$ with bilinear form $\langle x, x^* \rangle := x^*(x)$ is the exact same as the dual system $\langle X^*, X \rangle$ with bilinear form $\langle x^*, x \rangle := x^*(x)$. This leads to the following definition:

Definition A.6.1. If X is a TVS, the *weak* topology* on X^* is $\sigma(X^*, X)$; i.e., it is the initial topology generated by $\{\langle x^*, \cdot \rangle : x^* \in X^*\}$.

Theorem A.6.1. *If X is a LCS, then $(X^*, \sigma(X^*, X))^* \cong X$.*

Proof. Theorem A.5.5 ■

Remark. Since X is the dual of $(X^*, \sigma(X^*, X))^*$, does this mean there is a third topology on X , namely the weak* topology? The answer is no. The topology $\sigma((X, \sigma(X^*, X)), X^*)$ is the weakest topology on X making the maps $\{\langle \cdot, x^* \rangle : x^* \in X^*\}$ continuous. But this is precisely $\sigma(X, X^*)$, hence the topologies are the same.

Definition A.6.2. Let X be a TVS. If $A \subseteq X$ is closed in $\sigma(X, X^*)$, we say A is *weakly closed*. The closure of A in $\sigma(X, X^*)$ is denoted $\text{cl}^* A$.

Proposition A.6.2. *If X is a LCS and $A \subseteq X$ is convex, then $\text{cl} A = \text{cl}^* A$.*

Proof. ■

Corollary A.6.3. *Let X be a LCS. A convex subset of X is closed if and only if it is weakly closed.*

Proof. ■

Definition A.6.3. Let X be a TVS. If $A \subseteq X$, the *polar* of A , denoted A° , is the subset of X^* defined by:

$$A^\circ := \{x^* \in X^* : |\langle a, x^* \rangle| \leq 1 \text{ for all } a \in A\}.$$

If $B \subseteq X$, the *prepolar* of B , denoted ${}^\circ B$, is the subset of X defined by:

$${}^\circ B := \{x \in X : |\langle x, b^* \rangle| \leq 1 \text{ for all } b^* \in B\}.$$

If $E \subseteq X$, the *bipolar* of E is the set ${}^\circ E^\circ$.

Theorem A.6.4 (Alaoglu's Theorem). *Let X be a TVS and $\mathcal{B}$ its local base. If $V \in \mathcal{B}$, then V° is compact in $\sigma(X^*, X)$; i.e., it is weak*-compact.*

Proof. ■

Theorem A.6.5 (Banach-Alaoglu). *If X is a normed linear space, then $\overline{B}_{X^*}(0, 1)$ is compact in $\sigma(X^*, X)$; i.e., it is weak*-compact.*

Proof. ■

§ A.7. The Stone-Čech Compactification

Chapter B

Complex Variables

§ B.1. Basic Definitions and Results

Definition B.1.1. Let $G \subset \mathbb{C}$ be open and $f : G \rightarrow \mathbb{C}$.

- (1) The function f is *differentiable at a point* $a \in G$ if

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

exists. The value of this limit is denoted $f'(a)$. If f is differentiable for each $a \in G$, then we simply say f is *differentiable*.

- (2) The function f is *analytic* if f is continuously differentiable.

Definition B.1.2. A *path* is a continuous function $\gamma : [a, b] \rightarrow \mathbb{C}$.

Definition B.1.3. Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a path.

- (1) If $\gamma'(t)$ exists for each $t \in [a, b]$ and $\gamma' : [a, b] \rightarrow \mathbb{C}$ is continuous, then γ is a *smooth path*.
- (2) If there is a partition of $[a, b]$ such that γ is smooth along each subinterval, then γ is *piecewise smooth*.
- (3) If γ is of bounded variation, then it is called a *rectifiable path*.
- (4) The set $\{\gamma(t) : a \leq t \leq b\}$ is called the *trace* of γ and is denoted by $\{\gamma\}$.
- (5) The path γ is said to be *closed* if $\gamma(a) = \gamma(b)$.

Theorem B.1.1. Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a rectifiable path, and let $f : [a, b] \rightarrow \mathbb{C}$ be continuous. There exists a unique $I \in \mathbb{C}$ such that, for every $\epsilon > 0$, there exists $\delta > 0$ with the following property: if $\{t_0 < t_1, \dots, < t_m\}$ is a partition of $[a, b]$ with $\max\{t_k - t_{k-1} : 1 \leq k \leq m\} < \delta$, then for every choice of points $x_k \in [t_{k-1}, t_k]$ we have:

$$\left| I - \sum_{k=1}^m f(x_k) (\gamma(t_k) - \gamma(t_{k-1})) \right| < \epsilon.$$

Proof. ■

Definition B.1.4. The number $I \in \mathbb{C}$ established in Theorem B.1.1 is called the *Riemann-Stieltjes integral* of f with respect to γ over $[a, b]$ and is denoted by:

$$\int_a^b f d\gamma.$$

Theorem B.1.2. If γ is a piecewise smooth path and $f : [a, b] \rightarrow \mathbb{C}$ is continuous then:

$$\int_a^b f \, d\gamma = \int_a^b f(t) \gamma'(t) \, dt.$$

Proof. ■

Definition B.1.5. If $\gamma : [a, b] \rightarrow \mathbb{C}$ is a rectifiable path and $f : \{\gamma\} \rightarrow \mathbb{C}$ is continuous, then the *line integral of f along γ* is:

$$\int_a^b f(\gamma(t)) \, d\gamma(t).$$

This line integral is also denoted by $\int_\gamma f = \int_\gamma f(z) \, dz$.

Example B.1.1. Let $a \in \mathbb{C}$ be arbitrary and consider the path $\gamma : [0, 1] \rightarrow \mathbb{C}$ given by $\gamma(t) := a + e^{2\pi i n t}$. Let $f : \mathbb{C} \setminus \{a\} \rightarrow \mathbb{C}$ be given by $f(z) := (z - a)^{-1}$. The line integral of f along γ is:

$$\begin{aligned} \int_\gamma f(z) \, dz &= \int_0^1 f(\gamma(t)) \, d\gamma(t) \\ &= \int_0^1 e^{-2\pi i n t} \, d\gamma(t). \end{aligned}$$

Since $e^{-2\pi i n t}$ is smooth, Theorem ?? gives:

$$\begin{aligned} \int_0^1 e^{-2\pi i n t} \, d\gamma(t) &= \int_0^1 e^{-2\pi i n t} \gamma'(t) \, dt \\ &= \int_0^1 e^{-2\pi i n t} (2\pi i n e^{2\pi i n t}) \, dt \\ &= 2\pi i n. \end{aligned}$$

Lemma B.1.3. If $\gamma : [a, b] \rightarrow \mathbb{C}$ is a rectifiable path and $\varphi : [c, d] \rightarrow [a, b]$ is continuous, non-decreasing, and surjective, then $\gamma \circ \varphi : [c, d] \rightarrow \mathbb{C}$ is a rectifiable path and $\{\gamma\} = \{\gamma \circ \varphi\}$.

Proof. ■

Proposition B.1.4. Let $\gamma : [a, b] \rightarrow \mathbb{C}$ and $\sigma : [c, d] \rightarrow \mathbb{C}$ be rectifiable paths. Define $\sigma \sim \gamma$ if there exists a function $\varphi : [c, d] \rightarrow [a, b]$ such that φ is continuous, strictly increasing, $\varphi(c) = a$ and $\varphi(d) = b$, and $\sigma = \gamma \circ \varphi$. Then $\sim$ is an equivalence relation.

Proof. ■

Definition B.1.6. Using the terminology of Proposition ??, two paths γ and σ are *equivalent* if $\gamma \sim \sigma$. The function φ is called a *change of parameter*. An equivalence class of paths is called a *curve*.

Remark. The trace of a curve is the trace of any of its representatives. If f is continuous on the trace of a curve, then the integral over f over the curve is the integral of f over any representative of the curve. A curve is smooth if and only if one of its representatives is smooth.

Lemma B.1.5. *If G is an open set in $\mathbb{C}$, $\gamma : [a, b] \rightarrow G$ is a rectifiable path, and $f : G \rightarrow \mathbb{C}$ is continuous, then for every $\epsilon > 0$ there is a polygonal path Γ in G such that $\Gamma(a) = \gamma(a)$, $\Gamma(b) = \gamma(b)$, and $\left| \int_{\gamma} f - \int_{\Gamma} f \right| < \epsilon$.*

Proof. ■

Theorem B.1.6. *Let $G \subseteq \mathbb{C}$ be open and let γ be a rectifiable path in G with initial and end points α and β respectively. If $f : G \rightarrow \mathbb{C}$ is a continuous function with primitive $F : G \rightarrow \mathbb{C}$, then:*

$$\int_{\gamma} f = F(\beta) - F(\alpha),$$

(Recall that F is a primitive of f when $F' = f$.)

Proof. ■

Corollary B.1.7. *Let $G \subseteq \mathbb{C}$ be open and let γ be a closed rectifiable path in G . If $f : G \rightarrow \mathbb{C}$ is a continuous function with primitive $F : G \rightarrow \mathbb{C}$, then $\int_{\gamma} f = 0$.*

§ B.2. Winding Numbers and Cauchy's Integral Formula

Proposition B.2.1. *If $\gamma : [0, 1] \rightarrow \mathbb{C}$ is a closed rectifiable curve and $a \notin \{\gamma\}$, then:*

$$\frac{1}{2\pi i} \int_{\gamma} \frac{dz}{z - a}$$

is an integer.

Proof. ■

Definition B.2.1. If γ is a closed rectifiable curve in $\mathbb{C}$, then for $a \notin \{\gamma\}$, the *index of γ with respect to the point a* is:

$$n(\gamma : a) := \frac{1}{2\pi i} \int_{\gamma} (z - a)^{-1} dz.$$

Cauchy's Theorem Stuff...

§ B.3. Banach-Valued Integration

Definition B.3.1. Let $G \subseteq \mathbb{C}$ be open and X a Banach space. If $f : G \rightarrow X$, define the *derivative of f at z_0* to be:

$$f'(z_0) := \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h},$$

provided the limit exists. If the derivative of f exists for all $z \in G$, we say that f is differentiable. The function f is called *analytic* if f is continuously differentiable on G .

Remark. Let X be a Banach space. For any $x \in X$, recall that the *evaluation map* is $\hat{\cdot} : X^* \rightarrow \mathbb{F}$ given by $\hat{x}(x^*) := x^*(x)$. We introduce the following notation for convenience:

- For $x^* \in X^*$, define $\langle \cdot, x^* \rangle : X \rightarrow \mathbb{F}$ by $x \mapsto x^*(x)$.
- For $x \in X$, define $\langle x, * \rangle : X^* \rightarrow \mathbb{F}$ by $x^* \mapsto \hat{x}(x^*) = x^*(x)$.

We call $\langle \cdot, \cdot \rangle$ the *canonical evaluation map*.

Proposition B.3.1. Let $G \subseteq \mathbb{C}$ be open and X a Banach space. If $f : G \rightarrow X$ is analytic and $x^* \in X^*$, then $z \mapsto \langle f(z), x^* \rangle$ is analytic on G and its derivative is $\langle f'(z), x^* \rangle$.

Proof. As $z \mapsto \langle f(z), x^* \rangle$ is the composition of continuous maps, it is surely continuous. If such a function is denoted by g , then for any $z_0 \in G$ we have:

$$\begin{aligned} g'(z_0) &= \lim_{h \rightarrow 0} \frac{g(z_0 + h) - g(z_0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\langle f(z_0 + h), x^* \rangle - \langle f(z_0), x^* \rangle}{h} \\ &= \lim_{h \rightarrow 0} \left\langle \frac{f(z_0 + h) - f(z_0)}{h}, x^* \right\rangle \\ &= \left\langle \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h}, x^* \right\rangle \\ &= \langle f'(z_0), x^* \rangle. \end{aligned}$$

Since $g'(z_0)$ exists for all $z_0 \in G$, g is differentiable. Since g' is the composition of continuous maps, it is continuous. Thus $z \mapsto \langle f(z), x^* \rangle$ is analytic. ■

Proposition B.3.2. Let $G \subseteq \mathbb{C}$ be open and X a Banach space. If $f : G \rightarrow X$ is a function such that $z \mapsto \langle f(z), x^* \rangle$ is analytic for each $x^* \in X^*$, then f is analytic.

Proof. Since $G \subseteq \mathbb{C}$ is open, choose $r > 0$ such that $\overline{B}(z, r) \subseteq G$. Let $0 < \delta < \frac{r}{2}$, and let s, t satisfy $0 < |s|, |t| < \delta$. Fix $x^* \in X^*$ with $\|x^*\| \leq 1$ and let $g := \langle f(\cdot), x^* \rangle$. Since $z, z - s, z - t \in B^\circ(z, r)$, we can apply Cauchy's Integral Formula to $g(z)$, $g(z - s)$, and $g(z - t)$ (respectively). With this, observe that:

$$\begin{aligned} \left| \frac{g(z + s) - g(z)}{s} - \frac{g(z + t) - g(z)}{t} \right| &= \left| \frac{1}{2\pi i} \int_{\partial \overline{B}(z, r)} \frac{g(w)(s - t)}{(w - z)(w - z - s)(w - z - t)} dw \right| \\ &\leq \frac{1}{2\pi} \int_{\partial \overline{B}(z, r)} \frac{|g(w)||s - t|}{|w - z||w - z - s||w - z - t|} dw \end{aligned}$$

Note that $|s - t| \leq 2\delta$ and $|w - z||w - z - s||w - z - t| \geq \frac{r^3}{8}$. Hence:

$$\begin{aligned} \left| \frac{g(z+s) - g(z)}{s} - \frac{g(z+t) - g(z)}{t} \right| &\leq \frac{8\delta}{\pi r^3} \int_{\partial \bar{B}(z,r)} |g(w)| dw \\ &\leq c\delta \sup_{w \in \partial \bar{B}(z,r)} |g(w)| \\ &= c\delta \sup_{w \in \partial \bar{B}(z,r)} |\langle f(w), x^* \rangle| \end{aligned}$$

Note that $\langle f(w), \cdot \rangle : X^* \rightarrow \mathbb{C}$ is continuous for all $x^* \in X^*$. Hence $|\langle f(w), x^* \rangle| = |x^*(f(w))| \leq \|x^*\| \|f(w)\| \leq \|f(w)\|$. The above inequality can be written as:

$$\left| \left\langle \frac{f(z+s) - f(z)}{s} - \frac{f(z+t) - f(z)}{t}, x^* \right\rangle \right| \leq c\delta \sup_{w \in \partial \bar{B}(z,r)} \|f(w)\|.$$

Since the right-hand side is independent of $x^* \in X^*$, we have:

$$\sup_{\|x^*\| \leq 1} \left| \left\langle \frac{f(z+s) - f(z)}{s} - \frac{f(z+t) - f(z)}{t}, x^* \right\rangle \right| \leq c\delta \sup_{w \in \partial \bar{B}(z,r)} \|f(w)\|.$$

By a corollary of the Hahn-Banach Theorem, this gives:

$$\left\| \frac{f(z+s) - f(z)}{s} - \frac{f(z+t) - f(z)}{t} \right\| \leq c\delta \sup_{w \in \partial \bar{B}(z,r)} \|f(w)\|.$$

Given any $\epsilon > 0$, choose $0 < \delta < \frac{r}{2}$ such that $\delta \leq \frac{\epsilon}{\sup_{w \in \partial \bar{B}(z,r)} \|f(w)\|}$. If $0 < |s|, |t| < \delta$, then $\left\| \frac{f(z+s) - f(z)}{s} - \frac{f(z+t) - f(z)}{t} \right\| < \epsilon$. Since X is complete, by the Cauchy Criterion for limits $f'(z) = \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h} = 0$. Since $z \in G$ was arbitrary, f is analytic. ■

Theorem B.3.3. Let $G \subseteq \mathbb{C}$ be open and X a Banach space. Let γ be a rectifiable curve in G and f a continuous function defined in a neighborhood of $\{\gamma\}$ with values in X . There exists a unique $I \in X$ such that, for every $\epsilon > 0$, there exists $\delta > 0$ with the following property: if $\{t_0 < t_1 < \dots < t_m\}$ is a partition of $[0, 1]$ with $\max\{t_k - t_{k-1} : 1 \leq k \leq m\} < \delta$, then for every choice of points $x_k \in [t_{k-1}, t_k]$ we have:

$$\left\| I - \sum_{k=1}^m f(\gamma(x_k))(\gamma(t_k) - \gamma(t_{k-1})) \right\| < \epsilon.$$

Proof. ■

Definition B.3.2. The element $I \in X$ established in Theorem ?? is called the *line integral of f along γ* and is denoted by:

$$\int_0^1 f(\gamma(t)) d\gamma(t).$$

This line integral is also denoted by $\int_\gamma f = \int_\gamma f(z) dz$.

Proposition B.3.4. *Let $G \subseteq \mathbb{C}$ be open and γ a rectifiable curve in G . Suppose f is a continuous function defined in a neighborhood of $\{\gamma\}$ with values in X . For every $x^* \in X^*$, we have:*

$$\left\langle \int_{\gamma} f, x^* \right\rangle = \int_{\gamma} \langle f(\cdot), x^* \rangle.$$

Theorem B.3.5 (Cauchy's Theorem). *If X is a Banach space, G is an open subset of $\mathbb{C}$, $f : G \rightarrow X$ is an analytic function, and $\gamma_1, \dots, \gamma_m$ are closed rectifiable curves in G such that $\sum_{j=1}^m n(\gamma_j; a) = 0$ for all $a \in \mathbb{C} \setminus G$, then $\sum_{j=1}^m \int_{\gamma_j} f = 0$.*

Theorem B.3.6 (Cauchy's Integral Formula). *If X is a Banach space, G is an open subset of $\mathbb{C}$, $f : G \rightarrow X$ is analytic, γ is a closed rectifiable curve in G such that $n(\gamma; a) = 0$ for every $a \in \mathbb{C} \setminus G$, and $\lambda \in G \setminus \{\gamma\}$, then for every integer $k \geq 0$:*

$$n(\gamma; \lambda) f^{(k)}(\lambda) = \frac{k!}{2\pi i} \int_{\gamma} (z - \lambda)^{-(k+1)} f(z) dz.$$

Theorem B.3.7 (Runge's Theorem I). *Let X be a Banach space, $K \subset \mathbb{C}$ be compact and let E be a subset of $(\mathbb{C} \cup \{\infty\}) \setminus K$ that meets each component of $(\mathbb{C} \cup \{\infty\}) \setminus K$. If f is a Banach-valued analytic function in a neighborhood of K , then there are rational functions f_n whose only poles lie in E such that $f_n \rightarrow f$ uniformly on K .*

Theorem B.3.8 (Runge's Theorem II). *Let X be a Banach space, $G \subseteq \mathbb{C}$ open, and $E \subset (\mathbb{C} \cup \{\infty\}) \setminus G$ meet every component of $(\mathbb{C} \cup \{\infty\}) \setminus G$. If $f : G \rightarrow X$ is analytic, then there are rational functions $\{f_n\}$ whose only poles lie in E such that $f_n \rightarrow f$ uniformly on compact subsets of G .*